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Original Investigation

Comparative quantitative investigations on brains of wild cavies (*Cavia aperea*) and guinea pigs (*Cavia aperea* f. *porcellus*). A contribution to size changes of CNS structures due to domestication

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ABSTRACT

Intraspecific allometric calculations of the brain to body size relation revealed distinct differences between 127 (67; 60) ancestral wild cavies and 82 (37; 45) guinea pigs, their domesticated relatives. The dependency of both measures from one another remained the same in both animal groups but the brains of guinea pigs were by 14.22% smaller at any net body weight. Consistent with results in other species the domestication of *Cavia aperea* is also characterized by a decrease of brain size. Fresh tissue sizes of the five brain parts medulla oblongata, cerebellum, mesencephalon, diencephalon and telencephalon were determined for 6 cavies and 6 guinea pigs by the serial section method. Additionally the sizes of 16 endbrain structures and those of the optic tract, the lateral geniculate body and the cochlear nucleus were measured. Different decrease values resulted for all these structures concomitant with domestication as was calculated from the amount of total brain size decrease and average relative structure values in the wild as well as the domesticated brain. The size decrease of the entire telencephalon (–13.7%) was within the range of the mean overall reduction as similarly was the case for the total neocortex (–10.7%) whereas the total allocortex (–20.9%) clearly was more strongly affected. The size decrease of the olfactory bulb (–41.9%) was extreme and clearly higher than found for the secondary olfactory structures (around –11%). The primary nuclei of other sensory systems (vision, audition) were decreased to less extent (lateral geniculate: –18.1%; cochlear nucleus: –12.6%). Mass decreases of pure white matter parts were nearly twice as high in contrast to associated grey matter parts (neocortex white versus grey matter; tractus opticus versus lateral geniculate body). The relatively great decrease values found for the limbic structures hippocampus (–26.9%) and schizocortex (–25.9%) are especially notable since they are in good conformity with domestication effects in other mammalian species. The findings of this study are discussed with regard to results of similar investigations on wild and domesticated gerbils (*Meriones unguiculatus*), the encephalization of the wild form, the special and species-specific mode and duration of domestication and in connection with certain behavioral changes as resulted from comparative investigations in ethology, socio-biology, endocrinology and general physiology.

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Introduction

The domestication of animals is in general connected with a decrease of total brain size. This was ascertained through allometric comparisons of the brain size to body size relationship for a greater number of individuals of the wild progenitors versus the domesticated derivatives including different races, breeds or strains. This is valid for mammals (see reviews Herre and Roehrs 1990; Kruska 1980; Kruska 1988; Kruska 2005; Kruska 2007) and for birds as well

(Ebinger 1995). Since wild living individuals of the stem species and the domesticated progeny are still of the same species (Herre and Roehrs 1990) this effect is a remarkable zoological example for intraspecific variability and plasticity of brain size. However, the dimensions of the brain size decrease due to domestication vary considerably from species to species. In eutherian mammals they range from 0% decrease (laboratory mice, Nord 1963; Frick and Nord 1963) to 34% (pigs, Kruska 1970) or approximately 35% (European cattle Kruska unpubl. observ.). Intriguingly, the brains of species at a higher level of encephalization (Carnivora, Artiodactyla, Perissodactyla) clearly are more strongly diminished in size (from 16% to 34%) due to domestication than are those of less encephalized species (Rodentia, Lagomorpha from 0% to 15%).

The results on the few domesticated rodent and lagomorph species are especially striking. From interspecific allometric

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calculations for the brain to body size relationship different encephalizations resulted for the 64 rodent and 5 lagomorph species investigated. Some species had larger, other ones smaller brains at any given body weight than the average. The wild forms of the species *Oryctolagus cuniculus* (European rabbit), *Cavia aperea* (cavy) and *Meriones unguiculatus* (Mongolian gerbil) had indices of 98, 90 and 88, respectively. They are roughly on the same but slightly on a higher encephalization level compared to the other two wild forms *Mus musculus* (house mouse) and *Rattus norvegicus* (brown rat) with values of only 66 and 62, respectively (Kruska 1980; Kruska 2005; Leybold 2000; Kruska and Steffen 2009). These indices may characterize the interspecific differences in brain sizes that resulted from species specific evolution and lifestyle adaptation during long lasting phylogenetic events. With regard to domestication and the intensity of intraspecific brain size decrease, however, a rather similar effect was recorded just for these species until today. That is – with the exception of 0% in laboratory mice and 8.7% in albinotic laboratory Wistar rats (Kruska 1975a; Kruska 1975b) – decrease values resulted to amounts of 13.1% for rabbits (Fischer 1973), 13.4% for guinea pigs (Ebinger et al. 1984), 15% for laboratory gerbils (Leybold 2000) and 12% for totally pigmented laboratory DA rats (Kruska 1988; Kruska 2005; Kruska 2007). Thus, they are of rather similar intensity. Surprisingly, this is the case although the wild species with different encephalizations are very diverse in phylogeny and lifestyle and they became domesticated at different times, at different places, for very different reasons and under different conditions (Zeuner 1963; Mason 1984; Clutton-Brock 1987; Herre and Roehrs 1990).

It was also documented for several mammal species that diverse parts of the brain are decreased in size to different degrees due to domestication. Concerning the general intensity of diminution and the sequence from lowest to highest reduction value some convergent but also divergent effects were recognized from species to species. These results additionally may point to evolutionary as well as functional and other implications (Kruska 1988; Kruska 2005; Kruska 2007). Therefore it is of interest to investigate these issues in more detail and to extend our knowledge with results on another species, *Cavia aperea*.

Cavies are endemic to the wild of South America. They are distributed in several subspecies from Venezuela over the Andes to Brazil, Paraguay, Uruguay and Argentina except for the Amazon basin and southern parts of Chile and Argentina (Hueckinghaus 1962; Eisenberg 1989; Redford and Eisenberg 1992; Eisenberg and Redford 1999; Woods and Kilpatrick 2006). Guinea pigs are descendants of cavies. They were domesticated rather early during human history compared to the other rodent species. As ascertained from archeological records they already appeared in human deposits from pre-columbian times at high altitudinal locations of nowadays Peru and Bolivia. They probably were domesticated as early as 5000 B.C. but were definitely so by 2500 B.C. Thus, they live under the domestication influence for at least 4500 years (Wing 1977; Wing 1986; Sandweiss and Wing 1997; Reitz and Wing 1999). However, since the brain size is mainly genetically determined (Weidemann 1970a; Kruska 1973; Kruska 2005; Bedi and Bhide 1988; Leybold 2000) it additionally must be recognized that guinea pigs can reproduce up to five generations per year (Trillmich 2000) and concerning the length of the domestication period they thus have lived under this influence for much more generations than have the other domesticated lagomorphs and rodents and even than the other so-called classical domesticated forms (e.g., dog, sheep, goat, cattle, pig, etc.). These were domesticated at earlier times but have a turnover of only one or two generations per year (Herre and Roehrs 1990; Zeuner 1963). Guinea pigs are still used for meat production by Andean people but world-wide they mainly serve as laboratory animals for research or as pets (Mueller-Haye 1984; Morales 1995).

Intraspecific allometric brain size to body size calculations were already performed earlier of wild cavies originally caught in Southern Peru compared with commonly used guinea pigs from breeds in Germany (Ebinger et al. 1984). This study was done by use of the total body weight and on only a limited number of 25 wild versus 37 domesticated individuals. Brain sizes of all these individuals were not included but compared with those used here. Several brain parts of three male cavies were also already determined earlier in comparison with other caviomorph rodents (Weidemann 1970b). As regards a relative brain composition of *Cavia aperea* these results are in general conformity with the results obtained here, but the investigated individuals were from a zoological garden and thus of uncertain wild origin. For this reason these data are not used here. Similarly the relative values of some brain parts given for only one guinea pig (Pirlot and Bee de Speroni 1987) were useless since no absolute data of fresh tissue are reported.

The aim of this study therefore is to re-calculate the intraspecific allometric relationship of brain to body size on new material and to determine the degree of size differences of total brain and several brain parts between wild cavies and guinea pigs.

Material and methods

Animals

Altogether 127 (67 males, 60 females) wild cavies (*Cavia aperea*) and 82 (37 males, 45 females) guinea pigs (*Cavia aperea* f. *porcellus*) were bred under human care at the former Institut fuer Haustierkunde of the University at Kiel. Founder individuals of the wild cavy group were in 1996 obtained from a breeding colony of Prof. Dr. Norbert Sachser, University of Muenster, Germany. They were descendants of an earlier import of individuals originally caught in the wild at the vicinity of Lujan (northern Argentina) about 60 km west of Buenos Aires. Although these wild cavies of the subspecies *Cavia aperea pamparum* are not from the Andean region and thus from the distribution area of the subspecies *Cavia aperea tschudii* where the domestication once started (Wing 1977; Wing 1986; Mueller-Haye 1984) they nevertheless are of the same species *Cavia aperea* (see Hueckinghaus 1961; Hueckinghaus 1962; Kuenzl and Sachser 1999; Kuenzl et al. 2003; Kruska and Steffen 2013; in contrast to Spotorno et al. 2006; Spotorno et al. 2007; Dunnun and Salazar-Bravo 2010). No differences in brain size occur between different subspecies of a species relative to body size (Roehrs and Ebinger 1978; Ebinger 1980). Therefore these animals can in general serve as representative for the species wild type. All the individuals used here resemble the wild cavy type of *Cavia aperea* in general appearance, size, coloration and species characteristic configuration of the upper jaw M3 in occlusal view (Hueckinghaus 1961; Hueckinghaus 1962). Guinea pigs (*Cavia aperea* f. *porcellus*) were bought from local breeders without any documentation of specific breeds or strains. Altogether the used individuals of this colony resemble a great variability of domesticated forms and fancy breeds with respect to outer appearance, body weight, coat color, hair structure and other features. However, they all shared the characteristic upper M3 occlusal configuration with the investigated wild cavies.

All the individuals of cavies as well as guinea pigs were kept under similar conditions as described earlier (Kruska and Steffen 2013). No interbreeding was performed between the wild and the domesticated individuals and no special breeding aims were followed within both groups. Keeping all the individuals under similar conditions was done on purpose. It seemed important to exclude or at least to equalize possible environmental and modificatory effects on the growth and size of the body as well as the brain.

Preparation, histological procedures, and size determination of brain parts

From time to time and over the years several animals were removed from the colonies and sacrificed under general anesthesia. This is in accordance with the official German regulations for research on animals and the German animal welfare act. New breeding groups were established. The removed animals had reached different individual ages from six months to over three years, in most cases around one year. They all were in a reproductive state and thus adult.

The sacrificed individuals were immediately dissected, several body measures as well as body and organ weights were recorded. Body weights were documented as total body weight (TBW) and additionally as net body weight (NBW) which means total body weight minus weight of viscera and body fat. Brains were removed from the skulls (with the dura mater left), freshly weighed (fresh brain weight = FBW) and fixed by immersion in the fluid EFA (a mixture of 80 ml 80% ethanol, 10 ml of stock formaldehyde, and 10 ml of glacial acetic acid). After some weeks they were embedded in paraffin and stored until later usage.

Out of this large sample, brains of 6 (3 males, 3 females) cavies and of 6 (4 males, 2 females) guinea pigs were randomly selected for quantitative cytoarchitectonic investigations. These brains were serially sectioned (20 μ m thick) in the transverse plane and a series of about 250 equidistant sections of each brain was mounted on slides and stained with cresyl violet. The paper projection method was then used for volume calculations of total brain tissue and specific brain structures as outlined by Stephan (1960; Stephan et al. 1981) and also described earlier (e.g., Kruska 1993; Kruska and Steffen 2009). The total brain volumes as determined through the serial section method were always less than the actual fresh volumes of the total brain due to individual shrinkage caused by fixation and histological processing (Table 1). Consequently, the calculated volumes of the serial sections were corrected by an individual conversion factor for each brain to result in actual fresh tissue volumes (see Stephan 1960, specific gravity of brain substance = 1.036 g/cm³).

Altogether sizes of the following brain parts were measured: medulla oblongata, cerebellum, mesencephalon, diencephalon, telencephalon, within the telencephalon the total neocortex, with the grey matter and the white matter part, the corpus striatum, and the total allocortex. Furthermore the olfactory allocortex was measured with its structures olfactory bulb, regio retrobulbaris, tuberculum olfactorium, regio praepiriformis, nucleus amygdaloideus and basal nuclei and the non-olfactory part with the septum telencephali, the hippocampus and the schizocortex (=regio entorhinalis, prae- and parasubiculum). Additionally the sizes of the tractus opticus and the corpus geniculatum laterale were measured as representative structures of the visual system and likewise the nucleus cochlearis as the primary brain part for the ascending acoustic pathway (Tables 2 and 3).

The borders and delineations of these structures are in accordance with those used by e.g., Stephan et al. (1991) and previous studies on other mammalian brains (Kruska 1970; Kruska 1975a; Kruska 1975b; Kruska 1980; Kruska 1993; Kruska 1996; Kruska and Schott 1977; Kruska and Stephan 1973; Plogmann and Kruska 1990; Leybold 2000; Kruska and Steffen 2009).

Allometric and comparative calculations

Comparisons of brain part sizes between or within species are very often accomplished using the relative values (percentages of total brain sizes). A greater percentage of a certain brain region in one compared with other species or individuals is then interpreted as an enlargement or progression, a smaller value as a diminution

or regression due to evolution or/and life style adaptation (e.g., Campi and Krubitzer 2010; Krubitzer et al. 2011; Corfield et al. 2012). When doing this, however, it is excluded that differences of total brain sizes may be valid to occur between the species or groups compared. In other words: it is then totally ignored that the 100% brain size of one species in respect of body size may not be the same compared with the total brain size of another species since both species may be encephalized to different degrees. Additionally it is not realized that comparisons of relative values cannot give any insight in or information about either the direction or the dimension of size alterations. For these reasons also in this study the brains of the wild cavies and those of the guinea pigs were at first investigated and tested for any significant differences in total size.

This was done using the allometric method since brain size comparisons of differently sized animals must be made using the relationship of brain weight to body weight. This is described by the linear function: $\log y = a \log x + \log b$ in a double log plot of data sets (y = brain weight; x = body weight; a = slope; b = intersection of allometric line with the y -axis in $x=0$). No sexual dimorphic differences in brain size were established and therefore intraspecific but intersexual allometries (main axis) were calculated on the total samples of wild cavy ($n = 127$) and guinea pig ($n = 82$) data set for the relationships FBW to TBW and for FBW to NBW. In each of these two relationships the allometric lines of wild cavies and guinea pigs were then tested for differences in slope (F_a – value) and in position (F_b – value) on a 99.9% significance level (Rempe 1962). This was accomplished following the one-way ANOVA F -test procedure for the within-group and between-group variability using the program DIVA (Plogmann 1989). Since no significant differences in slope resulted for the wild versus the domesticated groups common slope values were calculated being valid for both groups and differences of total brain sizes were established independent of TBW or of NBW using the statistically different b values of the two respective parallel running lines.

The calculations of size differences for the measured brain parts (DBP) are then based on the average difference of total brain size (DTB; in this case as resulted from the FBW/NBW relationship) and the average relative values (RV) – as established and described earlier in detail (Kruska and Stephan 1973) – following the formula: $DBP = (RV_{\text{dom}}/RV_{\text{wild}}) \times (100 - DTB) - 100$ (where DBP = percentage difference of brain part size of the domesticated form compared to the wild form; RV = relative value; dom = domesticated form; wild = wild form; DTB = percentage difference of total brain size as resulted from allometric calculations on larger sample sets).

The relative brain part values needed for such calculations can be obtained in two ways, both leading to similar results. On the one hand the individual relative values may be used (Tables 2 and 3) to result in average values in form of arithmetical means. In this study, however, as preferred previously the geometrical means of absolute individual brain parts were tested for variance by the geometrical variability coefficient (GVC) at first and then used to obtain the average relative values (mean of total brain volume = 100%). These data and the resulting DBP values are listed in Table 4 for the diverse structures.

Results

At first it seems important to take issue to the shrinkage values obtained for the investigated individual brains as a consequence of fixation and histological processing. From Table 1 it becomes especially clear that the fresh brain sizes individually shrank to dimensions from 24.8% to 42.5%. Very remarkably this happened although all the brains were from individuals of the same species,

Table 1
Percentual shrinkage of total brain volume due to fixation and histological processing as calculated from the sizes of the serial section volumes compared to the fresh brain volumes.

Wild cavies, prep. no./sex	12,269 male	19,263 male	19,267 male	19,256 female	19,255 female	19,257 female
Shrinkage	35.55%	34.58%	32.59%	38.15%	39.61%	24.83%
Guinea pigs, prep. no./sex	18,477 male	18,206 male	18,205 male	18,719 male	18,731 female	18,733 female
Shrinkage	41.93%	40.97%	42.47%	40.44%	41.75%	36.97%

of the same age class, of approximately the same size and although they were treated the same during the histological processing. This is in accordance with earlier observations and experience with other brains where such shrinkage values even may exceed 60% (Kruska 1980; Kruska and Steffen 2009). Therefore, doubts become evident again concerning the validity of any inter- or intraspecific comparison when weights of total brains or brain parts are used taken after fixation or else absolute sizes of neocortical surface as well as grey matter thickness attained from serial sections.

However, fresh volume sizes and relative values of the investigated brains and brain parts of wild cavies are given in Table 2 those of guinea pigs in Table 3. Especially the relative values document a low individual variability within both groups. From these values it results that some brain regions of the guinea pigs are slightly larger in relative size compared with the wild cavies (e.g., total neocortex, neocortex grey matter) whereas some others are smaller (e.g., total allocortex, neocortex white matter, non-olfactory allocortex). Therefore it might be assumed that size increases as well as decreases occur due to domestication. However, as mentioned above such comparisons of relative sizes are extremely doubtful since it is not recognized that total brain sizes are different.

Calculating the FBW/TBW and the FBW/NBW relationships for both groups resulted in intraspecific intersexual allometric lines for

wild cavies and the derived guinea pigs. In both relationships these allometric lines of wild cavies were identical with those of guinea pigs in slope ($F_a = 1.2629$ and $F_a = 1.0798$, both values <10.8260 at $p = 0.001$) but significantly different in position ($F_b = 191.8876$ and $F_b = 194.9386$, both values >6.9080 at $p = 0.001$). Thus, in either of both relationships the guinea pig lines run parallel but transposed to that of the wild cavy lines at greater body but smaller brain sizes. Thus, the following is evident also within this species: (1) Intraspecific slope values for the brain size to body size relationship are in general smaller than for interspecific comparisons; (2) The relationship between brain and body weight remains unchanged and not affected in the domesticated form; (3) On average guinea pigs have smaller brains than their wild progenitors at any given body size. The following lines were established for both groups for the FBW/TBW relation:

Wild cavy : $\log y = 0.17 \log x + 0.2435-$

Guinea pig : $\log y = 0.17 \log x + 0.1645$

As resulted from the b values the brains of guinea pigs are on an average by 16.62% smaller compared to wild cavies at any total body weight. Additionally, an allometric comparison with the data of Ebinger et al. (1984) resulted as follows: The 25 cavies originally

Table 2
Brain weights (in g) and absolute volumes (in mm³) of diverse brain structures in wild cavies and their relative sizes expressed as percentages of pure brain volumes (in parentheses).

Prep. no./sex	19,269 male	19,263 male	19,267 male	19,256 female	19,255 female	19,257 female
Total brain weight	5.08	4.99	4.80	5.12	4.98	4.66
Total brain volume	4903.475	4816.602	4633.205	4942.085	4806.950	4498.069
Rest tissue	107.483	121.655	138.197	138.622	172.125	128.561
Ventricle	41.518	48.533	32.920	53.624	26.900	38.823
Pure brain tissue	4754.474 (100.00)	4646.414 (100.00)	4462.088 (100.00)	4749.839 (100.00)	4607.925 (100.00)	4330.685 (100.00)
Medulla oblongata	451.855 (9.50)	439.797 (9.47)	400.874 (8.98)	411.301 (8.66)	453.304 (9.84)	418.884 (9.67)
Cerebellum	678.657 (14.27)	757.298 (16.30)	546.788 (12.25)	609.631 (12.83)	741.572 (16.09)	562.940 (13.00)
Mesencephalon	448.557 (9.44)	431.672 (9.29)	396.245 (8.88)	406.549 (8.56)	406.411 (8.82)	360.479 (8.32)
Diencephalon	440.215 (9.26)	403.452 (8.68)	422.822 (9.48)	442.676 (9.32)	411.318 (8.93)	399.643 (9.23)
Telencephalon	2735.190 (57.53)	2614.195 (56.26)	2695.359 (60.41)	2879.682 (60.63)	2595.320 (56.32)	2588.739 (59.78)
Neocortex (total)	1624.662 (34.17)	1543.886 (33.23)	1577.608 (35.36)	1694.457 (35.67)	1515.679 (32.89)	1538.464 (35.52)
Corpus striatum	192.655 (4.05)	178.314 (3.83)	179.004 (4.01)	183.877 (3.88)	187.757 (4.07)	184.923 (4.27)
Allocortex (total)	917.873 (19.31)	891.995 (19.21)	938.747 (21.04)	1001.348 (21.08)	891.884 (19.36)	865.352 (19.99)
Neocortex (grey matter)	1335.971 (28.10)	1270.216 (27.34)	1295.555 (29.03)	1352.751 (28.48)	1250.131 (27.13)	1271.467 (29.36)
Neocortex (white matter)	288.691 (6.07)	273.670 (5.89)	282.053 (6.33)	341.706 (7.19)	265.548 (5.76)	266.997 (6.16)
Olfactory allocortex	399.665 (8.41)	382.070 (8.22)	394.189 (8.84)	407.689 (8.58)	364.242 (7.91)	347.706 (8.03)
Bulbus olfactorius	100.498 (2.11)	107.757 (2.32)	103.562 (2.32)	107.247 (2.26)	101.056 (2.19)	92.290 (2.13)
Regio retrobulbaris	10.477 (0.22)	13.898 (0.30)	11.830 (0.27)	12.931 (0.27)	8.725 (0.19)	11.578 (0.27)
Tuberculum olfactorium	38.802 (0.82)	32.927 (0.71)	36.864 (0.83)	39.933 (0.84)	32.716 (0.71)	34.907 (0.81)
Regio praepiriformis	114.468 (2.41)	113.530 (2.44)	113.680 (2.55)	118.655 (2.50)	107.237 (2.33)	95.696 (2.21)
Nucleus amygaloides	104.572 (2.20)	86.805 (1.87)	100.819 (2.26)	94.696 (1.99)	87.062 (1.89)	87.693 (2.02)
Basal nuclei	30.848 (0.65)	27.153 (0.58)	27.434 (0.61)	34.227 (0.72)	27.446 (0.60)	25.542 (0.59)
Non-olfactory allocortex	518.208 (10.90)	509.925 (10.98)	544.558 (12.20)	593.659 (12.50)	527.642 (11.45)	517.646 (11.96)
Septum telencephali	52.190 (1.10)	51.528 (1.12)	50.237 (1.13)	60.278 (1.27)	57.980 (1.26)	49.722 (1.15)
Hippocampus	338.358 (7.12)	345.295 (7.43)	357.152 (8.00)	398.752 (8.40)	346.976 (7.53)	351.965 (8.13)
Schizocortex	127.660 (2.68)	113.102 (2.43)	137.169 (3.07)	134.629 (2.83)	122.686 (2.66)	115.959 (2.68)
Tractus opticus	23.864 (0.50)	26.085 (0.56)	23.490 (0.53)	23.008 (0.48)	23.992 (0.52)	20.093 (0.46)
Corpus geniculatum laterale	15.327 (0.34)	17.746 (0.38)	14.745 (0.33)	16.354 (0.34)	11.087 (0.24)	14.985 (0.35)
Nucleus cochlearis	17.849 (0.38)	18.387 (0.40)	16.632 (0.37)	18.255 (0.38)	21.266 (0.46)	17.369 (0.40)

Table 3

Brain weights (in g) and absolute volumes (in mm³) of diverse brain structures in guinea pigs and their relative sizes expressed as percentages of pure brain volumes (in parentheses).

Prep. no./sex	18,477 male	18,206 male	18,205 male	18,719 male	18,731 female	18,733 female
Total brain weight	4.84	4.82	4.47	4.09	4.84	4.77
Total brain volume	4671.815	4652.251	4314.672	3947.876	4671.815	4604.247
Rest tissue	165.471	143.203	143.372	91.767	90.727	82.183
Ventricle	50.773	53.085	23.896	36.781	31.801	36.686
Pure brain tissue	4455.571 (100.00)	4455.963 (100.00)	4147.404 (100.00)	3819.328 (100.00)	4549.287 (100.00)	4485.378 (100.00)
Medulla oblongata	436.318 (9.79)	426.523 (9.57)	381.287 (9.19)	347.566 (9.10)	412.670 (9.07)	436.452 (9.73)
Cerebellum	649.233 (14.57)	663.754 (14.90)	557.211 (13.44)	553.579 (14.49)	604.602 (13.29)	618.083 (13.78)
Mesencephalon	337.434 (7.57)	370.353 (8.31)	341.115 (8.23)	326.018 (8.54)	398.265 (8.75)	358.948 (8.00)
Diencephalon	425.330 (9.55)	407.183 (9.14)	403.625 (9.73)	364.844 (9.55)	418.844 (9.21)	452.818 (10.10)
Telencephalon	2607.256 (58.52)	2588.150 (58.08)	2464.166 (59.41)	2227.321 (58.32)	2714.906 (59.68)	2619.077 (58.39)
Neocortex (total)	1579.800 (35.46)	1560.214 (35.01)	1507.137 (36.34)	1352.926 (35.42)	1688.282 (37.11)	1613.815 (35.98)
Corpus striatum	178.622 (4.01)	207.399 (4.65)	208.651 (5.03)	182.793 (4.79)	223.733 (4.92)	198.175 (4.42)
Allocortex (total)	848.834 (19.05)	820.537 (18.42)	748.378 (18.04)	691.602 (18.11)	802.891 (17.65)	807.087 (17.99)
Neocortex (grey matter)	1314.779 (29.51)	1302.201 (29.22)	1273.379 (30.70)	1154.157 (30.22)	1426.762 (31.36)	1340.830 (29.89)
Neocortex (white matter)	265.021 (5.95)	258.013 (5.79)	233.758 (5.64)	198.769 (5.20)	261.520 (5.75)	272.985 (6.09)
Olfactory allocortex	388.877 (8.73)	362.537 (8.14)	325.362 (7.84)	307.998 (8.07)	348.318 (7.66)	353.010 (7.87)
Bulbus olfactorius	77.576 (1.74)	64.608 (1.45)	61.298 (1.48)	60.373 (1.58)	57.990 (1.27)	68.695 (1.53)
Regio retrobulbaris	9.656 (0.22)	16.872 (0.38)	11.774 (0.28)	10.774 (0.28)	13.282 (0.29)	7.373 (0.16)
Tuberculum olfactorium	38.954 (0.87)	33.742 (0.76)	31.515 (0.76)	33.438 (0.88)	34.795 (0.77)	34.528 (0.77)
Regio praepiriformis	127.850 (2.87)	121.805 (2.73)	104.759 (2.53)	102.727 (2.69)	118.413 (2.60)	120.487 (2.69)
Nucleus amygaloides	99.882 (2.24)	94.029 (2.11)	90.214 (2.17)	78.950 (2.07)	93.534 (2.06)	87.398 (1.95)
Basal nuclei	34.959 (0.79)	31.481 (0.71)	25.802 (0.62)	21.736 (0.57)	30.304 (0.67)	34.529 (0.77)
Non-olfactory allocortex	459.957 (10.32)	458.000 (10.28)	423.016 (10.20)	383.604 (10.04)	454.573 (9.99)	454.077 (10.12)
Septum telencephali	53.437 (1.20)	61.313 (1.38)	48.657 (1.17)	42.354 (1.11)	47.515 (1.04)	51.433 (1.15)
Hippocampus	283.832 (6.37)	289.080 (6.49)	285.358 (6.88)	255.984 (6.70)	301.178 (6.62)	294.026 (6.55)
Schizocortex	122.688 (2.75)	107.607 (2.41)	89.001 (2.15)	85.266 (2.23)	105.880 (2.33)	108.618 (2.42)
Tractus opticus	17.479 (0.39)	20.987 (0.47)	14.719 (0.35)	14.490 (0.38)	15.714 (0.35)	17.444 (0.39)
Corpus geniculatum laterale	13.317 (0.30)	12.757 (0.29)	13.679 (0.33)	12.447 (0.33)	13.469 (0.30)	14.387 (0.32)
Nucleus cochlearis	19.644 (0.44)	17.695 (0.40)	18.008 (0.43)	15.605 (0.41)	17.584 (0.39)	16.365 (0.36)

Table 4

Data on the geometrical means (GM, in mm³) and the geometrical variability coefficients (GVC) of the measured brain parts, their percentages (relative values = RV) in relation to total brain volumes (=100%) of *n* = 6 wild cavies (wild) and *n* = 6 guinea pigs (dom) and the percentage size differences (DBP) of the diverse structures between wild cavies and guinea pigs based on a total brain size decrease of 14.22% due to domestication.

	GM wild	GVC wild	RV wild	GM dom	GVC dom	RV dom	DBP
Total brain weight	4.935	3.3		4.629	6.4		
Total brain volume	4764.310	3.3	100.0000	4468.893	6.4	100.0000	−14.22%
Pure brain tissue	4588.809	3.4	96.3163	4311.218	6.4	96.4717	−14.08%
Medulla oblongata	428.845	4.8	9.0012	405.415	8.6	9.0719	−13.55%
Cerebellum	644.318	13.6	13.5238	606.317	7.2	13.5675	−13.94%
Mesencephalon	407.380	7.1	8.5507	354.568	6.9	7.9341	−20.41%
Diencephalon	419.662	4.1	8.8085	411.244	6.7	9.2024	−10.38%
Telencephalon	2682.872	3.8	56.3119	2531.629	6.6	56.6500	−13.70%
Neocortex (total)	1581.248	3.9	33.1894	1546.678	7.2	34.6099	−10.55%
Corpus striatum	184.374	2.7	3.8699	199.297	8.1	4.4597	−1.15%
Allocortex (total)	916.854	4.8	19.2442	784.693	7.0	17.5590	−21.73%
Neocortex (grey matter)	1295.388	2.9	27.1894	1299.272	6.6	29.0737	−8.28%
Neocortex (white matter)	285.365	8.9	5.9896	246.945	11.4	5.5259	−20.86%
Olfactory allocortex	382.032	5.7	8.0186	346.737	7.8	7.7589	−17.00%
Bulbus olfactorius	101.930	5.3	2.1394	64.774	10.2	1.4494	−41.89%
Regio retrobulbaris	11.447	16.2	0.2403	11.246	29.4	0.2517	−10.15%
Tuberculum olfactorium	35.917	7.9	0.7539	34.427	6.6	0.7704	−12.34%
Regio praepiriformis	110.281	7.3	2.3147	115.638	8.4	2.5876	−4.11%
Nucleus amygaloides	93.347	7.7	1.9593	90.427	7.6	2.0235	−11.41%
Basal nuclei	28.635	10.2	0.6010	29.404	18.3	0.6580	−6.08%
Non-olfactory allocortex	534.564	5.2	11.2202	437.925	6.8	9.7994	−25.08%
Septum telencephali	53.506	7.6	1.1231	50.455	12.0	1.1290	−13.77%
Hippocampus	355.877	5.5	7.4696	284.512	5.3	6.3665	−26.89%
Schizocortex	124.882	7.4	2.6212	102.400	13.2	2.2914	−25.01%
Tractus opticus	23.351	8.1	0.4901	16.669	13.5	0.3730	−34.72%
Corpus geniculatum laterale	14.890	15.7	0.3125	13.329	4.8	0.2983	−18.12%
Nucleus cochlearis	18.239	7.9	0.3828	17.438	7.6	0.3902	−12.56%

caught in the wild of Peru had on average 3.4% smaller brains at any given total body weight than the 127 individuals investigated here. This value is within the normal intraspecific variability and documents in general similar brain sizes for the wild form. Therefore it is proved that the wild cavies investigated here although kept under human care for several generations still show no decreases of total brain size eventually caused by a “sneaking” domestication. On the other hand, the 37 guinea pigs of Ebinger et al. (1984) had on average by 1.9% larger brains at any total body weight compared with the 82 individuals of this study. Also this value is within the normal variability. The results for the FBW/NBW allometric lines are:

Wild cavy : $\log y = 0.18 \log x + 0.2595$ –

Guinea pig : $\log y = 0.18 \log x + 0.1929$

From these two lines resulted that guinea pigs have on an average by 14.22% smaller brains than their wild progenitor at any net body weight. This is illustrated in Fig. 1.

The decrease of total brain size to an amount of 14.22% was then used to calculate size differences for the diverse structures by use of their geometrical means. These are summarized in Table 4 as are values of the GVC. When comparing the DBP values it again is ascertained that the brain does not change homogeneously concomitant with domestication but in a mosaic mode. In general all the investigated structures are decreased in size, some to greater degree than the total brain others are less influenced. Altogether the diverse structures show decrease values from only 1.15% (corpus striatum) to 41.89% (bulbus olfactorius). With regard to the five classical brain parts the medulla oblongata (–13.55%), the cerebellum (–13.94%) and the telencephalon (–13.70%) are similarly diminished at approximately the same degree as is valid for the total brain, whereas the mesencephalon (–20.41%) is the brain part with the greatest and the diencephalon (–10.38%) with the lowest DBP value.

Of further interest are the endbrain parts. Here, the total allocortex (–21.73%) is clearly more affected than the total neocortex (–10.55%) and the corpus striatum (–1.15%) hardly ever. Very surprisingly the grey matter part of the neocortex is only slightly decreased (–8.28%) and very clearly to a lower degree compared with the associated white matter (–20.86%).

Also the size of the total allocortex has changed with the olfactory part (–17.00%) less prominent compared to the non-olfactory (–25.08%) structures. However, the olfactory bulb (–41.89%) as the primary region of olfaction is more strongly reduced compared with the secondary structures of this functional system. The values for the hippocampus (–26.89%) and the schizocortex (–25.01%) on the other hand indicate similar but prominent size decreases of limbic structures from the wild to the domesticated animal.

Additionally primary regions of two other sensory systems are of interest, namely those of vision and of audition. Here, the DBP value for the optic tract (–34.72%) is very prominent and nearly twice as high as that calculated for the directly associated lateral geniculate body (–18.12%). The cochlear nucleus (–12.56%) shows a clear but not exceptionally high size decrease.

Discussion

Evolutionary implications and species peculiarities in brain composition

In general the results of this study clearly indicate again that guinea pigs have smaller brains than their ancestor, the wild cavy. Thus a decrease of total brain size is assured concomitant with domestication. Only small differences can be recognized as regards the amount of this size diminution compared with other but similar

investigations on the same species. For example, earlier allometric studies calculated a decrease to an amount of 13.4% the wild animal's brain size when related to total body size (Ebinger et al. 1984). Calculations on braincase volumes of 93 wild cavies and 76 guinea pigs led to a slightly higher decrease value of 16.2% in relation to total body size (Kruska and Steffen 2013) which is the same as resulted here (16.62%) from the enlarged sample and based on the brain weight to total body weight relation. Measures of brain cavity sizes are not always consistent with the actual brain sizes. Very often differences occur between these two measures and normally these are greater in larger skulls (probably because of a thicker dura mater) than in smaller ones. This was especially evident when contrasting dwarf versus giant breeds of dogs (Roehrs and Ebinger 1978; Roehrs and Ebinger 1983). Yet, in *Cavia aperea* braincase size and brain size are reduced to same degree due to domestication which may be a proof for the fact that in this species the measure of braincase size directly equals brain size. The same was true for wild and ranch mink (Kruska and Sidorovich 2003), especially when contrasting both measures for the same individuals (Kruska pers. obs.).

However, a slightly smaller average difference of brain size between wild cavies and guinea pigs (14.22%) resulted when related to net body weights. This again may indicate that total body weights in relation to net body weights are not the same in the wild compared with the domesticated form; even so, although all the individuals of both groups were kept under similar conditions. The same resulted from investigations on wild and ranch mink after a life under human care and generally identical environmental influences (Kruska 1996). It is also important to realize that the wild cavies used here still represent the wild type although kept and bred under human care for several generations. As at last resulted here brain sizes of the used *Cavia aperea pamparum* from captivity were not different from those of *Cavia aperea tschudii* (Ebinger et al. 1984) that were caught in the wild. For these reasons the total brain size decrease to an amount of 14.22% concomitant with domestication was evaluated as the most reliable value for *Cavia aperea*. It was used for further calculations to determine size differences of diverse brain parts.

Certain evolutionary implications still remain valid as regards encephalization of the wild form versus intensity of brain size changes caused by domestication. For example, recent rodent and lagomorph species have on average brains on a lower encephalization plateau. In relation to body weight their brains are only about half the size compared with those of recent artiodactyl and carnivore species. Regarding this it is striking that concomitant with domestication the intraspecific brain size decreases of the rodent and lagomorph domesticates do not exceed about half the intensity as in general was determined for the domestications of the artiodactyl and carnivore orders (Kruska 1988; Kruska 2005; Kruska 2007). This phenomenon may be connected with the fact that the wild ancestors of artiodactyl and carnivore domesticates clearly have greater endbrains (about 70–80%) relative to total brain size in contrast to those of the less encephalized rodent and lagomorph orders (only about 54–58%). And just the telencephalon resulted for the artiodactyl and carnivore domesticates to be the prominent part with the greatest size decrease due to domestication when the five classical brain parts are considered. Apparently this is not the case on the lower encephalization level at least not in *Cavia aperea* as resulted here and not in *Meriones unguiculatus* (Leybold 2000). In guinea pigs the mesencephalon is diminished to greatest extend and in laboratory gerbils it is the cerebellum.

The mosaic and very arbitrary mode of size changes due to domestication with several brain subdivisions involved to different degrees becomes especially impressive when contrasting the decrease values of endbrain structures between the two rodent species *Cavia aperea* and *Meriones unguiculatus* (Leybold 2000). Here there are clear differences and only little conformities

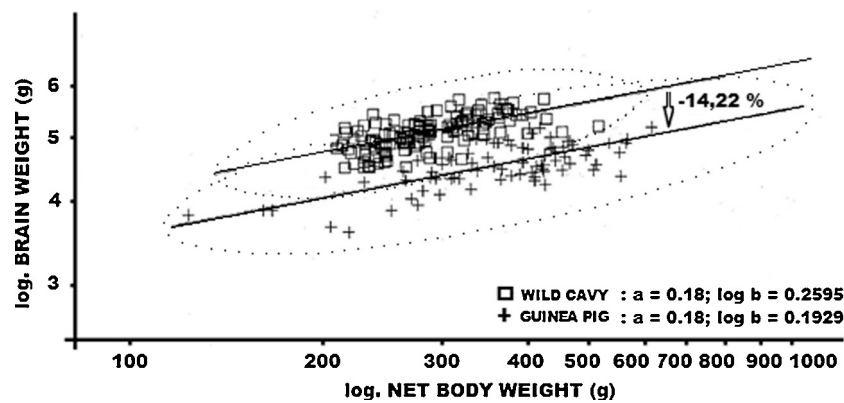


Fig. 1. Intraspecific relation of brain to net body weight in wild cavies (*Cavia aperea*) and guinea pigs (*Cavia aperea* f. *porcellus*). Both allometric lines run significantly assured parallel (same a -value) but transposed (different b -values) to one another indicating a brain size decrease of 14.22% concomitant with domestication.

although both species are similarly encephalized (with indices of 90 and 88, respectively) and the decrease values of pure brain tissue in the domesticates are the same (14% and 15%, respectively). The telencephalon is certainly a little more strongly reduced in guinea pigs (14%) in contrast to laboratory gerbils (11%) but within the endbrain the prominent parts, the neocortex and the allocortex participate very differently. In *Cavia aperea* the size decrease of total neocortex (11%) is only half as large compared with the total allocortex (22%), whereas this is directly inverse in *Meriones unguiculatus* (with values of 14% and 7%, respectively). The reduction of total neocortex size is mainly caused by the white matter part. This tissue is diminished to same degree (21%) in both species in contrast to the grey matter volume. This is only 8% smaller in guinea pigs but 13% in laboratory gerbils. These corresponding results in two rodent species, namely that the white substance of neocortex is more strongly affected due to domestication than the associated grey matter part was also valid for other species on the higher encephalization level (Kruska 1980; Kruska 2007). An explanation for this may be that with each neuron loss in ascending or/and descending pathways the mass of axoplasm is clearly involved to less extent in contrast to the larger mass of the surrounding myelin sheath.

Differences are also valid for the two main parts of the total allocortex, namely the olfactory and the non-olfactory regions. Both these regions are only little affected in gerbils (9% and 4.5%, respectively) but clearly more strongly in guinea pigs (17% and 25%, respectively). Additionally for the latter the extremely high value of 42% found for the olfactory bulb most probably indicates an extreme loss of olfactory performance although the secondary olfactory regions are decreased to less intensity of altogether around 11%. In laboratory gerbils the olfactory bulb is reduced by only 8% and the secondary olfactory regions by 8–14%. However, the two main structures of the non-olfactory part that describe the center of the limbic system, namely the hippocampus and the schizocortex are especially decreased in guinea pigs (27% and 25%, respectively) compared with the low values in gerbils (1% and 7%, respectively) that document nearly no changes at all. Very notably such extreme values for the size decrease of the hippocampus that even exceed those for the total brain and the neocortex size were found earlier also for the so-called classical domestic forms like pigs, sheep and dogs (Kruska 1988; Kruska 2005; Kruska 2007). In connection with the main function of this limbic center they were described as the most characteristic attribute of the domestication effect in the brain. As resulted here this is also the case for *Cavia aperea*.

Besides the olfactory bulb the primary centers of two other sensory systems are also smaller in guinea pigs in contrast to wild cavies. With regard to the visual system “the white matter” of the tractus opticus (35%) is decreased in size about twice as much as

the associated “grey matter” of the lateral geniculate nucleus (18%) similarly as resulted for these two tissue parts of the neocortex. The nucleus cochlearis, at least, with a decrease dimension (13%) comparable with total brain size diminution may indicate the smallest loss for the auditory system compared with the primary regions of the other two sensory systems. Quantitative investigations on the skulls of wild cavies and guinea pigs resulted also in significant reductions of length and breadth of the bony auditory bulla (Kruska and Steffen 2013) and thus confirm these changes due to domestication.

Size changes of CNS tissue and behavioral changes due to domestication

Following the assumption that the described size decreases of central nervous tissue caused by domestication are indeed connected with gradual changes of certain functions and consequently of behavioral pattern, then results of comparative ethological and physiological research between wild cavies and guinea pigs are of interest. Especially size decreases of grey matter parts may roughly indicate decreases of neuron numbers and therewith quantitative or/and qualitative loss of special performances. Such investigations that compare sensory abilities of the two *Cavia aperea* types are rare. Although both forms were erroneously treated as different species (*Cavia aperea*, *Cavia porcellus*) by Beauchamp and colleagues olfactory communication was investigated for example, by use of urine preference tests. As a general result the wild cavy males were more strongly attracted by female and male wild urine whereas the guinea pig males preferred urine of the domesticated subjects (Beauchamp et al. 1979). This was in accordance with investigations on scent from the perineal glands concerning their chemical composition and attraction to the animals. These glands produce scent with volatiles evidently group specific different in composition. Due to domestication they obviously have diverged. Thus, wild males investigated wild scents more preferably than domestic secretions. The converse resulted for the domestic males (Wellington et al. 1979). However, it is hardly possible to conclude whether more central nervous mass is needed to differentiate between the wild and the domestic scent. A loss of olfactory performance due to domestication as a consequence of the size decrease of olfactory brain structures is not proved by these experiments. Especially not, since cross-rearing experiments clearly demonstrated that imprinting plays an important role in determining odor preferences (Beauchamp and Wellington 1981).

Other sensory abilities were also investigated as for example, regards taste. Here, males of both *Cavia aperea* forms were tested for preferences between glucose solutions of different concentrations and de-ionized water. As a result all the animals were attracted

by the glucose solution but a broader range of concentrations was preferred by the wild subjects. This was assumed to indicate a greater ability of taste discrimination. Thus, it was concluded that the wild type may possess more receptor cells than the domestic form (Jacobs and Beauchamp 1977). This may also be valid for the size of certain central nervous structures although no white substances or nuclei of gustatory quality were specially measured here.

Unfortunately, no special physiological or ethological investigations of other sensory abilities (visual, auditory) are known for *Cavia aperea*. In any case, the evaluation of results of such eventual comparisons may also bear some uncertainties since working experimentally with wild individuals often presents problems even after taming or habituation. These animals are constantly attentive, shy and aware under any man-made experimental condition or experiment design (see Steffen et al. 2001), most probably as a consequence of their greater brain size, larger limbic regions and other structural composition. After all wild cavies also showed a clearly higher stress responsiveness compared with guinea pigs (Kuenzl et al. 2003). Because of that they may not “concentrate” on the particular tasks and test procedures the same whereas domesticated forms may act rather “relaxed” and with better results (Kruska 1988). Evidently this turned out to be the case with regard to results of certain test procedures with the aim to evaluate learning abilities. For example, laboratory rats (Boice 1972) as well as domesticated gerbils (Stuerner and Wetzel 2006) were better learners than their wild counterparts. The same resulted for guinea pigs when spatial learning abilities were tested in a special design of water maze experiments (Lewejohann et al. 2010).

However, from general comparative observations it appears that the repertoire of behavior patterns is similar in wild cavies and guinea pigs. Thus, also in this species no new behavioral elements have developed and none got lost concomitant with domestication (Rood 1972; Stahnke 1987; Hendrichs and Stahnke 1988). This is in accordance with results on other species and in most cases the behavioral changes from the wild to the domesticated forms are to a certain degree characterized by a dissociation of action chains (Lorenz 1959) which again is in agreement with the differently sized decreases of the brain subdivisions (Kruska 1988; Kruska 2007). Distinct differences occurred, however, with respect to the frequencies with which specific behavioral elements were displayed. Wild cavies, both males and females displayed significantly higher levels of curved body posture, head thrust, attack lunge, chase and bite than the domesticated relatives. According to behavioral investigations under standardized as well as non-standardized conditions guinea pigs always displayed lower levels of intraspecific aggressive behavior and higher levels of sociopositive behavior than their wild counterparts. Thus, the process of domestication has led to typical traits-reduced aggressiveness with the consequence of increased tolerance against conspecifics (Sachser 1998; Kuenzl and Sachser 1999; Asher et al. 2004). Similarly to other domesticated forms (Kruska 2007) very probably this is connected with the evidently large size decrease of limbic structures. Guinea pigs are also less attentive to common environmental stimuli and they show reduced explorative behaviors (Sachser 1998) which again may be the result of size decreased sensory brain structures. On the other hand, overt courtship behavior is evidently expressed more frequently in guinea pigs (Sachser 1998). Therefore, the domestication of *Cavia aperea* is also connected with hypersexuality which is commonly known also from other species although no brain region is increased in size. In this regard some results of experimental studies on rats and cats seem of interest for an explanation (Richter 1954; Schreiner and King 1956; Green et al. 1957). These have elucidated the inhibitory character of certain allocortical structures, such as the prepiriform cortex and deeper parts of the amygdaloid complex. Removal and destruction

of these regions resulted in hypersexual behavior of the animals tested. Thus it seems that the size decrease of these nuclear masses due to domestication may be responsible for a behavioral increase (Kruska 1988).

Genetics and time dependency of brain size changes due to domestication

Altogether also for *Cavia aperea* the described quantitative changes of brain size and structural composition concomitant with domestication must be evaluated as a result of human interference and activities. They are, however, no modificatory effects eventually caused by a life in captivity and under restricted and impoverished environments. At last all the wild cavy and guinea pig individuals investigated here were kept under similar conditions. Yet, brain sizes remained larger in the wild cavies significantly distinguished from those of guinea pigs. On the contrary, brain size and structural composition are strictly speaking genetically determined. This resulted from interbreeding experiments of wild progenitors with domesticated descendants. For example, hybridism of wolves with poodles led in the first generation to individuals with brains at the same body weight on average intermediary in size between those of the parents. Only bastards of the second generation resulted in some individuals with large brains of wolf size, smaller ones of poodle size as well as sisters with brain sizes in between (Weidemann 1970a). The situation reminds of Mendelian rules. Just the same, also 20 individuals of the first offspring generation from a wild with laboratory gerbil interbreeding had brains on average only 7% smaller than their wild parents at any body weight in contrast to the 15% smaller brains their laboratory parents had (Leybold 2000). Even bastards of interspecific crossings between horses and donkeys resulted in offspring brain sizes intermediary to those of the parents (Kruska 1973).

Bearing all this in mind the following scenario may be assumed for the domestication in general and for *Cavia aperea* also. Very probably already early breeders at ancient times selected wild caught individuals suitable for a life under conditions of domestication. Very certainly the breeding aims were manifold during the following generations under human care but changes of brain size and composition surely were never possible directly to estimate and to aim at. Such changes only are imaginable to result indirectly as a consequence of consciously or/and unconsciously evaluated ethological and behavioral peculiarities of the breeding individuals. Consequently the full adaptation to the “ecological niche” of domestication including the brain size decrease was reached step by step and only over several generations until accomplished.

Therefore, it also seems of interest to know how many “steps” there might be needed to change an animal’s brain from the wild to the domesticated state. In other words: How fast does the brain size really change? In this respect, even today only little is known for sure and the discussion is still open (Bedi and Bhide 1988). For longer times in the past there was assumed that changes from a wild to a domesticated animal including brain size decrease are rather long-enduring effects (Herre and Roehrs 1990) although happening very much faster than those brain size changes that occurred during phylogenetic events (Kruska 1982; Kruska 1987). However, it was also noticed earlier (Hueckinghaus 1961) and again assured recently (Spotorno et al. 2007) that mummies of guinea pigs from pre-columbian times already had smaller brain-cases compared to wild cavies although it remained unknown to what extent and after how many generations. Mongolian gerbils were bred under laboratory conditions since 1935 but for unknown numbers of generations per year (Leybold 2000; Stuerner et al. 2003). Thus this species was domesticated for much shorter times in contrast to guinea pigs. Nevertheless, a similar decrease of total brain size was already reached 65 years later. Anyway, this may

point to rather early appearing brain size decreases during the course of domestication. The same holds true for dogs (Kruska 2005; Kruska 2007) but especially convincing for ranch mink. The domestication of this species started around 1866 and already after 70–80 years and generations later on the braincase was altered in size to dimensions characteristic for nowadays ranch mink (Kruska and Sidorovich 2003).

With regard to behavioral changes concomitant with domestication it is of interest that long-term breeding of wild caught cavies without any selective aims but only rearing the animals in captivity under human care did not result in significant changes of behavior and hormonal stress responses after a period of 30 generations (Kuenzl et al. 2003). The animals still remained wild but clearly different to guinea pigs. This is also valid for the brain size as resulted from this study. On the other hand, strict and rigorous selection of calm and aggressive attributes in domestic silver foxes was connected with destabilizing effects in this species (Belyaev 1979; Trut 1999). Especially the selection of tame and docile individuals adapted to the special environments of domestication led to silver foxes with nearly a dog-like behavior including some other traits of domesticated mammals already after little more than 50 generations (Trut et al. 2009). Unfortunately no brains were investigated of these animals. Nevertheless, it can be assumed that the size and structural changes of the central nervous system and behavior due to domestication are rather rapidly established if aimed at and they tend to remain stable further on (Kruska 2005). In any case, they are intraspecific short time effects of minor dimensions when contrasted with long-enduring and larger dimensioned interspecific events concomitant with the origin of new species following evolutionary adaptive radiations.

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